A-A-53861
17 March 1989

COMMERCIAL ITEM DESCRIPTION

INDICATOR, COMBUSTIBLE GAS, PORTABLE (Dual Range)

The General Services Administration has authorized the use of this commercial item description.

This commercial item description covers a portable, dual range combustible gas indicator for the detection and measurement of specific combustible gases or vapors.

Salient characteristics.

Shall be a portable, battery operated dual range combustible gas indicator for detecting and measuring the concentration of specific combustible gases or vapors in air. Shall also be suitable for the detection of gasoline vapor.

Shall be supplied with a dual scale meter movement that is calibrated on hexane gas.

Shall be supplied with correlation curves for interpreting meter readings for the following gases or vapors: Acetone, Butanol, Benzol, Butyl Acetate, Ethyl Acetate, Ethyl Alcohol, Ethyl Ether, Hexane, Isopropyl Alcohol, Methyl Alcohol, Methyl Ethyl Ketone, Toluol, and Xylol.

Shall have the dual scale meter supplied with a selector switch for changing from one scale to another. The major meter scale calibration range shall be from 0 to 100% of a lower flammable limit (L.F.L.) for a combustible gas or vapor. In addition, the major meter scale shall be calibrated to read 100% (i.e. representing "explosive") at 95% of the lower explosive limit of the hexane gas calibration standard. The minor meter scale calibration scale range shall be 0 to 10% of the lower flammable limit (L.F.L.) of a combustible gas or vapor. The correlation curves to be supplied shall be used to interpret the minor scale readings in terms of the parts per million (p.p.m.) concentration of a combustible gas or vapor in air.

AMSC N/A FSC 6665

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Shall be supplied with a two position switch for controlling the dual scale meter readings, in accordance with the following:

Provides a functional 1:1 position for the major scale readings.

Provides a functional 1/10 L.F.L. position for the minor scale readings.

Shall be designed to be electrically operated by type C or type D flashlight batteries (not required to be supplied with the combustible gas indicator).

Shall also include the following:

A rigid instrument housing with a neck strap.

An internal Wheatstone bridge electrical circuitry with a dial controlled zero adjustment potentiometer.

Two replaceable and interchangeable platinum wire filaments for use as resistance legs in the Wheatstone bridge electrical circuitry (see NOTES).

An internal chamber to secure one of the platinum wire filaments being used as the analyzer filament.

An external detachable drying chamber with an inlet fitting for a sampling hose.

A five (5) foot length of sampling hose with a suitable fitting at one end for attachment to the inlet of the drying chamber.

An external detachable rubber aspirator bulb assembly.

Workmanship. The combustible gas indicator shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One indicator, as specified, constitutes one unit.

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Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Inspection and Tests. Inspections and tests shall be conducted in accordance with commercial practice to determine compliance with specification requirements. Sampling for inspections and test shall also be conducted in accordance with commercial practice.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

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Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U. S. Food and Drug Administration to be under its jurisdiction, the offeror/ contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number (NSN) is covered by this document:

NSN 6665-00-542-1442 INDICATOR, COMBUSTIBLE GAS, PORTABLE, Dual Range

The following National Stock Number (NSN) covers a replacement combustible gas indicator filament for the combustible gas indicator.

NSN 6665-01-181-3172 FILAMENT, COMBUSTIBLE GAS INDICATOR

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Ordering data. (Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order).

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD - MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

PHS

Project No. 6665-0575

Location:
CID/INDICATOR/COMBUST/GAS/S31-16

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-53861	INDICATOR, COMBUSTABLE GAS, PORTABLE (Dual Range)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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